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Differential pair inner-side impedance compensation

Abstract

An information handling system includes first and second differential pairs on a printed circuit board. The first differential pair includes first and second traces, and first and second sets of impedance compensation traces. The first impedance compensation traces are routed only on an inner-side of the first trace. The second impedance compensation traces are routed only on an inner-side of the second trace, and the first and second impedance compensation traces are substantially aligned. The second differential pair includes third and fourth traces and third and fourth sets of impedance compensation traces. The third set of impedance compensation traces are routed only on an inner-side of the third trace. The fourth impedance compensation traces are routed only on an inner-side of the fourth trace, and the third and fourth impedance compensation traces are substantially aligned.

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Background/Summary

FIELD OF THE DISCLOSURE

(1) The present disclosure generally relates to information handling systems, and more particularly relates to differential pair inner-side impedance compensation.

BACKGROUND

(2) As the value and use of information continues to increase, individuals and businesses seek additional ways to process and store information. One option is an information handling system. An information handling system generally processes, compiles, stores, or communicates information or data for business, personal, or other purposes. Technology and information handling needs and requirements can vary between different applications. Thus information handling systems can also vary regarding what information is handled, how the information is handled, how much information is processed, stored, or communicated, and how quickly and efficiently the information can be processed, stored, or communicated. The variations in information handling systems allow information handling systems to be general or configured for a specific user or specific use such as financial transaction processing, airline reservations, enterprise data storage, or global communications. In addition, information handling systems can include a variety of

hardware and software resources that can be configured to process, store, and communicate information and can include one or more computer systems, graphics interface systems, data storage systems, networking systems, and mobile communication systems. Information handling systems can also implement various virtualized architectures. Data and voice communications among information handling systems may be via networks that are wired, wireless, or some combination.

SUMMARY

(3) An information handling system includes first and second differential pairs on a printed circuit board. The first differential pair includes first and second traces, and first and second sets of impedance compensation traces. The first impedance compensation traces may be routed only on an inner-side of the first trace. The second impedance compensation traces may be routed only on an inner-side of the second trace, and the first and second impedance compensation traces may be substantially aligned. The second differential pair includes third and fourth traces and third and fourth sets of impedance compensation traces. The third set of impedance compensation traces may be routed only on an inner-side of the third trace. The fourth impedance compensation traces may be routed only on an inner-side of the fourth trace, and the third and fourth impedance compensation traces may be substantially aligned.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) It will be appreciated that for simplicity and clarity of illustration, elements illustrated in the Figures are not necessarily drawn to scale. For example, the dimensions of some elements may be exaggerated relative to other elements. Embodiments incorporating teachings of the present disclosure are shown and described with respect to the drawings herein, in which:

(2) FIG. **1** is a diagram of differential pairs with dual-side impedance compensation according to prior art;

(3) FIG. **2** is a diagram of differential pairs with inner-side impedance compensation according to at least one embodiment of the present disclosure;

(4) FIG. **3** is two impedance curves and each impedance curve is associated with the impedance of different respective differential pairs based a different impedance compensation scheme according to at least one embodiment of the present disclosure;

(5) FIG. **4** is two return loss curves and each return loss curve is associated with the return loss of a different respective differential pair based on different impedance compensation schemes according to at least one embodiment of the present disclosure;

(6) FIG. **5** is two crosstalk curves and each crosstalk curve is associated with the crosstalk between differential pairs based on different impedance compensation schemes according to at least one embodiment of the present disclosure; and

(7) FIG. **6** is a flow diagram of method for implementing an inner-side impedance compensation scheme to a differential pair according to at least one embodiment of the present disclosure; and

(8) FIG. **7** is a block diagram of a general information handling system according to an embodiment of the present disclosure;

(9) The use of the same reference symbols in different drawings indicates similar or identical items.

DETAILED DESCRIPTION OF THE DRAWINGS

(10) The following description in combination with the Figures is provided to assist in understanding the teachings disclosed herein. The description is focused on specific implementations and embodiments of the teachings, and is provided to assist in describing the teachings. This focus should not be interpreted as a limitation on the scope or applicability of the teachings.

(11) FIG. **1** illustrates a printed circuit board (PCB) **100** of an information handling system, such as information handling system **700** of FIG. **7**, according to prior art in the field. For purpose of this disclosure information handling system can include any instrumentality or aggregate of instrumentalities operable to compute, classify, process, transmit, receive, retrieve, originate, switch, store, display, manifest, detect, record, reproduce, handle, or utilize any form of information, intelligence, or data for business, scientific, control, entertainment, or other purposes. For example, an information handling system can be a personal computer, a laptop computer, a smart phone, a tablet device or other consumer electronic device, a network server, a network storage device, a switch, a router, or another network communication device, or any other suitable device and may vary in size, shape, performance, functionality, and price.
(12) PCB **100** includes differential pairs **102** and **104**. Differential pair **102** includes traces **110** and **112**, and differential pair **104** includes traces **114** and **116**. Trace **110** of differential pair **102** and trace **114** of differential pair **104** may be substantially straight based on routing requirements within a ball grid array (BGA) pin field. However, spacing or room within the BGA pin field may be limited. In this situation, a high-speed differential signal transmitted along traces **110** and **112** of differential pair **102** and a high-speed differential signal transmitted along traces **114** and **116** of differential pair **104** may both have a dynamic phase matching requirement to improve signal transmission quality along the differential pair. Based on the phase matching requirement, trace **112** of differential pair **102** and trace **116** of differential pair **104** may include phase tuning bumping areas **118** to provide phase skew tuning within the respective differential pair. Signal integrity of high-speed signals transmitted along differential pairs **102** and **104** may be improved via other compensation schemes, such as impedance compensation, beyond the phase skew tuning described above.
(13) Impedance compensation along differential pairs **102** and **104** of PCB **100** may be implemented by increasing the differential pair trace width in the phase tuning area of the differential pairs. As shown in FIG. **1**, the width of trace **110** of differential pair **102** is increased by both an outer-side compensation **120** and an inner-side compensation **122** in the phase tuning area of the differential pair **102**. As used herein, outer-side compensation refers to a side of a trace furthest from the other trace of the differential pair, and inner-side compensation refers to a side of a trace nearest to the other trace of the differential pair. For clarity, only a few outer-side compensations **120** and a few inner-side compensations **122** have been labeled. However, one of ordinary skill in the art will recognize trace **110** may include outer-side compensations **120** and inner-side compensations **122** at each of the phase tuning areas of differential pair **102**.
(14) As described above with respect to trace **110**, the width of trace **112** of differential pair **102** is increased by an outer-side compensation **124** and an inner-side compensation **126** in the phase tuning bumping area **118**. For clarity, only a few outer-side compensations **124** and a few inner-side compensations **126** have been labeled. However, one of ordinary skill in the art will recognize trace **112** may include outer-side compensation **124** and inner-side compensation **126** at each of the phase tuning bumping areas **118** of differential pair **102**. The combination of outer-side and inner-side compensation may reduce return loss within differential pair **102**.
(15) As shown in FIG. **1**, the width of trace **114** of differential pair **104** is increased by both an outer-side compensation **130** and an inner-side compensation **132** in the phase tuning area of the differential pair **104**. For clarity, only a few outer-side compensations **130** and a few inner-side compensations **132** have been labeled. However, one of ordinary skill in the art will recognize trace **114** may include outer-side compensation **130** and inner-side compensation **132** at each of the phase tuning areas of differential pair **104**.
(16) As described above with respect to trace **114**, the width of trace **116** of differential pair **104** is increased by an outer-side compensation **134** and an inner-side compensation **136** in the phase tuning bumping area **118**. For clarity, only a few outer-side compensations **134** and a few inner-side compensations **136** have been labeled. However, one of ordinary skill in the art will recognize trace

116 may include outer-side compensation **134** and inner-side compensation **136** at each of the phase tuning bumping areas **118** of differential pair **104**. The combination of outer-side and inner-side compensation may reduce return loss within differential pair **104**.

(17) Crosstalk between the differential signal on differential pair **102** and the differential signal on differential pair **104** may be created or changed by any suitable structural characteristic of the differential pairs on PCB **100**. One such structural characteristic is a distance between differential pairs **102** and **104**. As shown in FIG. **1**, there are two different distances **140** and **142** between differential pairs **102** and **104**. Distance **140** is the largest distance between trace **112** of differential pair **102** and trace **114** of differential pair **104**, and this distance may be located at any point or location other than phase tuning bumping area **118** of trace **112**. Distance **140** may be any suitable distance including, but not limited to, thirteen and eight-hundredths mils. Distance **142** is the smallest distance between trace **112** of differential pair **102** and trace **114** of differential pair **104**, and this distance may be located within phase tuning bumping area **118** of trace **112**. In particular, distance **142** is the distance between outer-side compensation **130** of trace **114** and outer-side compensation **124** on phase tuning bumping area **118** of trace **112**. Distance **142** may be any suitable distance including, but not limited to, nine mils. One of ordinary skill in the art would recognize that mil is a unit of measurement utilized in routing on PCBs, and one mil equals one-thousandth of an inch or two hundred fifty four ten-thousandths of a millimeter.

(18) In this situation, distance **142** may have the greatest impact on the crosstalk between differential pairs **102** and **104**. Thus, while the combination of outer-side and inner-side compensations on traces **110**, **112**, **114**, and **116** may reduce return loss within respective differential pairs **102** and **104**, outer-side compensation **124** on trace **112** and outer-side compensation **130** of trace **114** may increase crosstalk between differential pairs **102** and **104**. The width of outer-side compensation **124** on trace **112** and outer-side compensation **130** of trace **114** may reduce distance **142**, which in turn may increase crosstalk between differential pairs **102** and **104**.

(19) Traces **110** and **112** of differential pair **102** and traces **114** and **116** of differential pair **104** have widths or spacings that may be substantially similar between the differential pairs. For clarity and brevity the widths and spacings will be described with respect to traces **110** and **112** of differential pair **102**. Traces **110** and **112** may have an edge-to-edge width **144**, compensation widths **146** and **148**, and compensation spacing **150**. Edge-to-edge width **144** may be the distance from an outermost edge of outer-side compensation **120** of trace **110** to an outermost edge of outer-side compensation **124** of trace **112**. Thus, edge-to-edge width **144** may vary based on the widths of outer-side compensations **120** and **124**. Edge-to-edge width **244** may be any suitable distance including, but not limited to, fifteen and eight-hundredths mils.

(20) Compensation width **146** is the width of trace **112** plus the width of outer-side compensation **134** and inner-side compensation **136**. Thus compensation width **146** may vary based on different widths of outer-side compensation **134** and inner-side compensation **136**. As shown in FIG. **1**, compensation width **146** may be measured or determined at phase tuning bumping area **118** of trace **112**. Compensation width **148** is the width of trace **110** plus the width of outer-side compensation **120** and inner-side compensation **122**, and compensation width **148** may be measured or determined at phase tuning bumping area of trace **110**. Thus compensation width **148** may vary based on different widths of outer-side compensation **120** and inner-side compensation **122**. In differential pairs **102** and **104** on PCB **100**, compensation widths **146** and **148** may be the same width, such as four and six-tenths mils. However, in different layouts of differential pairs **102** and **104**, compensation widths **146** and **148** may be different widths.

(21) Compensation spacing **150** is the width between inner-side compensation **122** of trace **110** and inner-side compensation **126** of trace **112**. Thus compensation spacing **150** may vary based on different widths of inner-side compensation **122** and inner-side compensation **126**. Compensation spacing **150** may be any suitable distance between inner-side compensation **122** of trace **110** and

inner-side compensation **126** of trace **112**, such as six and six-tenths mils.

(22) As stated above, while the combination of outer-side and inner-side compensations on traces **110**, **112**, **114**, and **116** may reduce return loss within respective differential pair **104**, outer-side compensation **124** on trace **112** and outer-side compensation **130** of trace **114** may increase crosstalk between differential pairs **102** and **104**. The width of outer-side compensation **124** on trace **112** and outer-side compensation **130** of trace **114** may reduce distance **142**, which in turn may increase crosstalk between differential pairs **102** and **104**.

(23) The inventor of the current disclosure have found an impedance compensation scheme that can keep the same impedance characteristics and return loss performance as compared to differential pairs **102** and **104**, but may improve crosstalk performance as will be described with respect to FIG. **2** below.

(24) FIG. **2** illustrates a PCB **200** including differential pairs **202** and **204** according to at least one embodiment of the present disclosure. Differential pair **202** includes traces **210** and **212**, and differential pair **204** includes traces **214** and **216**. Traces **210**, **212**, **214**, and **216** may be understood to represent a signal traces that carry data signals between components of an information handling system, such as information handling system **700** of FIG. **7**, and may include portions that are routed on a top or bottom surface of PCB **200**, within one or more layers of the PCB, and within one or more vias between layers of the PCB, as needed or desired. While differential pair set **102** and **104** and differential pair set **202** and **204** are illustrated in separate figures, one of ordinary skill in the art would recognize that both differential pair sets may be routed on the same PCB without varying from the scope of this disclosure.

(25) Trace **210** of differential pair **202** and trace **214** of differential pair **204** may be substantially straight based on routing requirements within a BGA pin field. However, spacing or room within the BGA field may be limited. In this situation, a high-speed differential signal transmitted along traces **210** and **212** of differential pair **202** and a high-speed differential signal transmitted along traces **214** and **216** of differential pair **204** may both have a dynamic phase matching requirement to improve signal transmission quality along the differential pair. Based on the phase matching requirement, trace **212** of differential pair **202** and trace **216** of differential pair **204** may include phase tuning bumping areas **218** to provide phase skew tuning within the respective differential pair. Signal integrity of high-speed signals transmitted along differential pairs **202** and **204** may be improved via other compensation schemes, such as impedance compensation, beyond the phase skew tuning described above.

(26) Impedance compensation along differential pairs **202** and **204** of PCB **200** may be implemented by increasing the differential pair trace width in the phase tuning area of the differential pairs. As shown in FIG. **2**, the width of trace **210** of differential pair **202** may be increased only by inner-side compensation **222** in the phase tuning area of the differential pair **202**. In an example, the width of inner-side compensation **222** of trace **210** may be larger than the width of inner-side compensation **122** of trace **110** to provide a similar impedance compensation for trace **210** as compared to trace **110** of FIG. **1**. In certain examples, the increase width of inner-side compensation **222** as compared to inner-side compensation **122** may be any suitable amount up to the combined widths of outer-side compensation **120** and inner-side compensation **122** of trace **110** of FIG. **1**. For clarity, only a few inner-side compensations **222** have been labeled. However, one of ordinary skill in the art will recognize trace **210** may include inner-side compensations **222** at each of the phase tuning areas of differential pair **202**.

(27) As described above with respect to trace **210**, the width of trace **212** of differential pair **202** may be increased only by inner-side compensation **226** in the phase tuning bumping area **218**. In an example, the width of inner-side compensation **226** of trace **212** may be larger than the width of inner-side compensation **126** of trace **112** to provide a similar impedance compensation for trace **212** as compared to trace **112** of FIG. **1**. In certain examples, the increase width of inner-side compensation **226** as compared to inner-side compensation **126** may be any suitable amount up to

the combined widths of outer-side compensation **124** and inner-side compensation **126** of trace **112** of FIG. **1**. For clarity, only a few inner-side compensations **226** have been labeled. However, one of ordinary skill in the art will recognize trace **212** may include inner-side compensations **226** at each of the phase tuning bumping areas **218** of differential pair **202**. Inner-side compensation **222** of trace **210** and inner-side compensation **226** of trace **212** may reduce return loss within differential pair **202**. In certain examples, the widths of inner-side compensations **222** and **226** may vary depending on a desired impedance compensation for differential pair **202**. For example, lesser impedance compensation may be provided if the widths of inner-side compensations **222** and **226** are reduced, and greater impedance compensation may be provided if the widths of inner-side compensations **222** and **226** are increased.
(28) As stated above, inner-side compensations **222** of trace **210** and inner-side compensations **226** of trace **212** may maintain a substantially similar impedance compensation for differential pair **202** as compared to differential pair **102** as will be illustrated and described with respect to FIGS. **3**.
(29) FIG. **3** illustrates two impedance curves **302** and **304** and each impedance curve is associated with the impedance of different respective differential pairs, such as differential pair **102** of FIG. **1** and differential pair **202** of FIG. **2**, based a different impedance compensation scheme according to at least one embodiment of the present disclosure. As shown in FIG. **3**, the vertical axis indicates an impedance of a differential pair and the horizontal axis indicates an amount of time measured in picoseconds. In an example, the impedance compensation for a differential pair may be utilized to maintain an impedance variation for the traces of the differential pair to within a one ohm variation, such as between eighty-eight ohms and eighty-nine ohms as illustrated in FIG. **3**.
(30) Impedance curve **302** may be associated with either differential pair **102** or **104** of FIG. **1**. Impedance curve **302** may be controlled based on the impedance compensations added to the traces of a particular differential pair. As stated above, impedance compensation along differential pairs **102** and **104** of PCB **100** may be implemented by increasing the differential pair trace width in the phase tuning area of the differential pairs. As shown in FIG. **1**, the width of trace **110** of differential pair **102** is increased by both outer-side compensation **120** and inner-side compensation **122** in the phase tuning area of the differential pair **102**. Similarly, the width of trace **112** of differential pair **102** is increased by both outer-side compensation **124** and inner-side compensation **126** in the phase tuning area of the differential pair **102**. Thus outer-side compensations **120** and **124** and inner-side compensations **122** and **126** within differential pair **102** may cause the differential pair to have impedance curve **302** and an impedance variation less than a one ohm limited as shown in FIG. **3**.
(31) Impedance curve **304** may be associated with either differential pair **202** or **204** of FIG. **2**. Impedance curve **304** may be controlled based on the impedance compensations added to the traces of a particular differential pair. As stated above, impedance compensation along differential pairs **202** and **204** of PCB **200** may be implemented by increasing the differential pair trace width in the phase tuning area of the differential pairs. As shown in FIG. **2**, the width of trace **210** of differential pair **202** is increased by only inner-side compensation **222** in the phase tuning area of the differential pair **202**. Similarly, the width of trace **212** of differential pair **202** is increased by only by inner-side compensation **226** in the phase tuning area of the differential pair **202**. Thus inner-side compensations **222** and **226** within differential pair **202** may cause the differential pair to have impedance curve **304** and an impedance variation less than a one ohm limited as shown in FIG. **3**.
(32) Referring back to FIG. **2**, inner-side compensations **222** of trace **210** and inner-side compensations **226** of trace **212** may maintain a substantially similar return loss for differential pair **202** as compared to differential pair **102** as will be illustrated and described with respect to FIGS. **4**.
(33) FIG. **4** illustrates two return loss curves **402** and **404** and each return loss curve is associated with the return loss of a different respective differential pair, such as differential pair **102** of FIG. **1** and differential pair **202** of FIG. **2**, based on the different impedance compensation schemes of

FIGS. **1** and **2** according to at least one embodiment of the present disclosure. As shown in FIG. **4**, the vertical axis indicates an amount of return loss and the horizontal axis indicates a frequency in Gigahertz (GHz). In an example, the impedance compensation for a differential pair may be utilized to maintain a particular return loss for the differential pair at a given frequency.

(34) Return loss curve **402** may be associated with either differential pair **102** or **104** of FIG. **1**. Return loss curve **402** may be controlled based on the impedance compensations added to the traces of a particular differential pair. As stated above, impedance compensation along differential pairs **102** and **104** of PCB **100** may be implemented by increasing the differential pair trace width in the phase tuning area of the differential pairs. As shown in FIG. **1**, the width of trace **110** of differential pair **102** is increased by both outer-side compensation **120** and inner-side compensation **122** in the phase tuning area of the differential pair **102**. Similarly, the width of trace **112** of differential pair **102** is increased by both outer-side compensation **124** and inner-side compensation **126** in the phase tuning area of the differential pair **102**. Thus outer-side compensations **120** and **124** and inner-side compensations **122** and **126** within differential pair **102** may cause the differential pair to have return loss curve **402** with a particular return loss at a given frequency, such as a return loss substantially equal to −30 dB at 21.58 GHz as indicated by point **410** in FIG. **4**.

(35) Return loss curve **404** may be associated with either differential pair **202** or **204** of FIG. **2**. Return loss curve **404** may be controlled based on the impedance compensations added to the traces of a particular differential pair. As stated above, impedance compensation along differential pairs **202** and **204** of PCB **200** may be implemented by increasing the differential pair trace width in the phase tuning area of the differential pairs. As shown in FIG. **2**, the width of trace **210** of differential pair **202** is increased by only inner-side compensation **222** in the phase tuning area of the differential pair **202**. Similarly, the width of trace **212** of differential pair **202** is increased by only by inner-side compensation **226** in the phase tuning area of the differential pair **202**. Thus inner-side compensations **222** and **226** within differential pair **202** may cause the differential pair to have return loss curve **402** with a return loss substantially equal to the return loss of point **510** at the same frequency of return loss curve **502**, such as a return loss substantially equal to −30 dB at 21.58 GHz as indicated by point **420** in FIG. **4**.

(36) Referring back to FIG. **2**, the width of trace **214** of differential pair **204** may be increased only by inner-side compensation **232** in the phase tuning area of the differential pair **204**. In an example, the width of inner-side compensation **232** of trace **214** may be larger than the width of inner-side compensation **122** of trace **114** to provide a similar impedance compensation for trace **214** as compared to trace **114** of FIG. **1**. In certain examples, the increase width of inner-side compensation **232** as compared to inner-side compensation **132** may be any suitable amount up to the combined widths of outer-side compensation **130** and inner-side compensation **132** of trace **114** of FIG. **1**.

(37) The width of trace **216** of differential pair **204** is increased only by inner-side compensation **236** in the phase tuning bumping area **218**. In an example, the width of inner-side compensation **236** of trace **216** may be larger than the width of inner-side compensation **136** of trace **116** to provide a similar impedance compensation for trace **216** as compared to trace **116** of FIG. **1**. The increase width of inner-side compensation **236** as compared to inner-side compensation **136** may be any suitable amount up to the combined widths of outer-side compensation **134** and inner-side compensation **136** of trace **116** of FIG. **1**. For clarity, only a few inner-side compensations **232** and **236** have been labeled. However, one of ordinary skill in the art will recognize trace **214** may include inner-side compensations **232** at each of the phase tuning areas of differential pair **204** and trace **216** may include inner-side compensations **236** at each of the phase tuning bumping areas **218**. Inner-side compensation **232** of trace **214** and inner-side compensation **236** of trace **216** may reduce return loss within differential pair **204**. In certain examples, the widths of inner-side compensations **232** and **236** may vary depending on a desired impedance compensation for

differential pair **204**. For example, lesser impedance compensation may be provided if the widths of inner-side compensations **232** and **236** are reduced, and greater impedance compensation may be provided if the widths of inner-side compensations **232** and **236** are increased.

(38) Crosstalk between the differential signal on differential pair **202** and the differential signal on differential pair **204** may be created or changed by any suitable structural characteristic of the differential pairs on PCB **200**. One such structural characteristic is a distance between differential pairs **202** and **204**. As shown in FIG. **2**, there are two different distances **240** and **242** between differential pairs **202** and **204**. Distance **240** is the largest distance between trace **212** of differential pair **202** and trace **214** of differential pair **204**, and this distance may be located at any point or location other than phase tuning bumping area **218** of trace **212**. Distance **240** may be any suitable distance including, but not limited to, thirteen and eight-hundredths mils. In this example, distance **240** between differential pairs **202** and **204** may be substantially equal to distance **140** between differential pairs **102** and **104** on PCB **100** of FIG. **1**.

(39) Distance **242** is the smallest distance between trace **212** of differential pair **202** and trace **214** of differential pair **204**, and this distance may be located within phase tuning bumping area **218** of trace **212**. In particular, distance **242** is the distance between an outer edge of trace **214** and an outer edge of phase tuning bumping area **218** of trace **212**. In this example, the outer edges of traces **212** and **214** may be the edges of the traces themselves without an outer-side compensation trace being adding to the traces. Distance **242** may be any suitable distance including, but not limited to, ten mils. In this example, distance **240** between differential pairs **202** and **204** may be greater than distance **140** between differential pairs **102** and **104** on PCB **100** of FIG. **1**.

(40) In this situation, distance **242** may improve the far-end crosstalk (FEXT) performance of differential pairs **202** and **204** and compared to the crosstalk performance of differential pairs **102** and **204**. In an example, the combination of inner-side compensations **222** and **226** on respective traces **210** and **212** may reduce return loss within differential pair **202**. Similarly, the combination of inner-side compensations **232** and **236** on respective traces **214** and **216** may reduce return loss within differential pair **204**. Additionally, based on traces **210**, **212**, **214**, and **216** only including respective inner-side compensations **222**, **226**, **232**, and **236**, the FEXT between differential pairs **202** and **204** may be reduced. In this example, inner-side compensations **222**, **226**, **232**, and **236** do not affect distance **242** between differential pairs **202** and **204**, which in turn may reduce or suppress the FEXT between differential pairs **202** and **204** as compared to differential pairs **102** and **104** as will be illustrated and described with respect to FIG. **5**.

(41) FIG. **5** illustrates crosstalk curves **502** and **504** associated with the crosstalk between differential pairs based on different impedance compensation schemes according to at least one embodiment of the present disclosure. As shown in FIG. **4**, the vertical axis indicates an amount of loss for crosstalk between differential pairs and the horizontal axis indicates a frequency in Gigahertz (GHz). In an example, the impedance compensation for a differential pair may affect crosstalk between the differential pair at a given frequency.

(42) Crosstalk curve **502** may be associated with differential pairs **102** and **104** of FIG. **1**. Crosstalk curve **502** may be affected based on the impedance compensations added to the traces of the differential pairs. As stated above, impedance compensation along differential pairs **102** and **104** of PCB **100** may be implemented by increasing the differential pair trace width in the phase tuning area of the differential pairs. As shown in FIG. **1**, the width of trace **110** of differential pair **102** is increased by both outer-side compensation **120** and inner-side compensation **122** in the phase tuning area of the differential pair **102**. Similarly, the width of trace **112** of differential pair **102** is increased by both outer-side compensation **124** and inner-side compensation **126** in the phase tuning area of the differential pair **102**. Thus outer-side compensations **120** and **124** and inner-side compensations **122** and **126** within differential pair **102** may result in distance **142** between differential pairs **102** and **104**.

(43) Distance **142** may result in differential pairs **102** and **104** to have crosstalk curve **402** with a

particular crosstalk at respective frequencies. Differential pairs **102** and **104** may be utilized to provide a high-speed signal, such as a signal at either 16 GHz or 28 GHz. Differential pairs **102** and **104** may have a power sum far-end crosstalk (PSFEXT) suppression substantially equal to −42 dB at 16 GHz as indicated by point **510** in FIG. **5**. Similarly, differential pairs **102** and **104** may have a power sum far-end crosstalk (PSFEXT) suppression substantially equal to −38 dB at 28 GHz as indicated by point **512** in FIG. **5**.

(44) Crosstalk curve **504** may be associated with differential pairs **202** and **204** of FIG. **2**. Crosstalk curve **504** may be affected based on the impedance compensations added to the traces of a particular differential pair. As stated above, impedance compensation along differential pairs **202** and **204** of PCB **200** may be implemented by increasing the differential pair trace width in the phase tuning area of the differential pairs. As shown in FIG. **2**, the width of trace **210** of differential pair **202** is increased by only inner-side compensation **222** in the phase tuning area of the differential pair **202**. Similarly, the width of trace **212** of differential pair **202** is increased by only by inner-side compensation **226** in the phase tuning area of the differential pair **202**. Thus inner-side compensations **222** and **226** within differential pair **202** may result in distance **242** between differential pairs **202** and **204**.

(45) As stated above, distance **242** may be greater than distance **142** of differential pairs **102** and **104**, such that the FEXT performance of differential pairs **202** and **204** may be improved as compared to differential pairs **102** and **104**. In an example, distance **242** may result in differential pairs **202** and **204** to have crosstalk curve **504** with a particular FEXT at respective frequencies. Differential pairs **202** and **204** may be utilized to provide substantially similar high-speed signals as differential pairs **102** and **104**. As shown in FIG. **5**, crosstalk curve **504** has a greater PSFEXT loss or suppression for all frequencies of a signal as compared to crosstalk curve **502**. For example, crosstalk curve **504** indicates that differential pairs **202** and **204** may have a PSFEXT suppression substantially equal to −45 dB at 16 GHz as indicated by point **520** in FIG. **5**. This PSFEXT suppression may be substantially equal to a 3 dB PSFEXT suppression improvement for differential pairs **202** and **204** as compared to differential pairs **102** and **104**. Similarly, differential pairs **202** and **204** may have a PSFEXT suppression substantially equal to −41 dB at 28 GHz as indicated by point **514** in FIG. **5**. This PSFEXT suppression may be substantially equal to a 3 dB PSFEXT suppression improvement for differential pairs **202** and **204** as compared to differential pairs **102** and **104**.

(46) Traces **210** and **212** of differential pair **202** and traces **214** and **216** of differential pair **204** have widths or spacings that may be substantially similar between the differential pairs. For clarity and brevity the widths and spacings will be described with respect to traces **210** and **212** of differential pair **202**. Traces **210** and **212** may have an edge-to-edge width **244**, compensation widths **246** and **248**, and compensation spacing **250**. Edge-to-edge width **244** may be the distance from an outermost edge trace **210** to an outermost edge of trace **212**. Thus, edge-to-edge width **244** may vary based on the widths of traces **210** and **212**. Edge-to-edge width **244** may be any suitable distance including, but not limited to, fourteen and eight-hundredths mils. In an example, edge-to-edge width **244** of traces **210** and **212** and the edge-to-edge width of traces **214** and **214** may both be less than edge-to-edge width **144** of traces **110** and **112** and the edge-to-edge width of traces **114** and **116** of FIG. **1**. In this example, traces **210** and **212** of differential pair **202** and traces **214** and **216** of differential pair **204** may consume or cover a smaller amount space on PCB **200** as compared to the amount of space that differential pairs **102** and **104** consume or cover on PCB **100** of FIGS. **1**.

(47) Compensation width **246** is the width of trace **212** plus the width of inner-side compensation **236**. Thus compensation width **246** may vary based on different widths of inner-side compensation **236**. As shown in FIG. **2**, compensation width **246** may be measured or determined at phase tuning bumping area **218** of trace **212**. Compensation width **248** is the width of trace **210** plus inner-side compensation **222**, and compensation width **248** may be measured or determined at phase tuning

bumping area of trace **210**. Thus compensation width **248** may vary based on different widths of outer-side compensation **220** and inner-side compensation **222**. In differential pairs **202** and **204** on PCB **200**, compensation widths **246** and **248** may be the same width, such as four and four-tenths mils. However, in different layouts of differential pairs **202** and **204**, compensation widths **246** and **248** may be different widths.

(48) Compensation spacing **250** is the width between inner-side compensation **222** of trace **210** and inner-side compensation **226** of trace **212**. Thus compensation spacing **250** may vary based on different widths of inner-side compensation **222** and inner-side compensation **226**. Compensation spacing **250** may be any suitable distance between inner-side compensation **222** of trace **210** and inner-side compensation **226** of trace **212**, such as six mils.

(49) As stated above, inner-side compensations **222** of trace **210** and inner-side compensations **226** of trace **212** may maintain substantially similar impedance compensation and a substantially similar return loss for differential pair **202** as compared to differential pair **102**. Similarly, inner-side compensations **232** of trace **214** and inner-side compensations **236** of trace **216** may maintain substantially similar impedance compensation and a substantially similar return loss for differential pair **204** as compared to differential pair **104**. Additionally, traces **210**, **212**, **214**, and **216** having only respective inner-side compensations **222**, **226**, **232**, and **236** may improve the crosstalk performance for differential pairs **202** and **204** may be improved as compare to differential pairs **102** and **104** as will be illustrated and described with respect to FIG. **6**.

(50) FIG. **6** is a flow diagram of method **600** for implementing an inner-side impedance compensation scheme to a differential pair according to at least one embodiment of the present disclosure, starting a block **602**. It will be readily appreciated that not every method step set forth in this flow diagram is always necessary, and that certain steps of the methods may be combined, performed simultaneously, in a different order, or perhaps omitted, without varying from the scope of the disclosure. FIG. **6** may be employed in whole, or in part, any other type of controller, device, module, processor, or any combination thereof, operable to employ all, or portions of, the method of FIG. **6**.

(51) At block **604**, a first trace of a differential pair is provided or routed on a PCB. In an example, the first trace may be any suitable length and may be provided in any suitable direction on the PCB. For example, the direction of the first trace on the PCB may be based on routing requirements within a BGA pin field of the PCB. At block **606**, a second trace of the differential pair is provided or routed on a PCB. In an example, may be routed a particular distance from the first trace to create the differential pair of the first and second traces. In certain examples, spacing or room within the BGA pin field may be limited. In this situation, a high-speed differential signal transmitted along the first and second traces of the differential pair may have a dynamic phase matching requirement to improve signal transmission quality along the differential pair. Based on the phase matching requirement, the second trace of the differential pair may include phase tuning bumping areas to provide phase skew tuning within the respective differential pair.

(52) At block **608**, a first set of impedance compensation traces may be provided or routed on an inner-side of the first trace. In an example, the first impedance compensation traces may be located at any suitable locations along the first trace. For example, the first impedance compensation traces may be located on the inner-side of the first trace at the phase tuning bumping areas of the differential pair.

(53) At block **610**, a second set of impedance compensation traces may be provided or routed on an inner-side of the first trace, and the flow ends at block **612**. In an example, the second impedance compensation traces may be located at any suitable locations along the second trace. For example, the second impedance compensation traces may be located on the inner-side of the phase tuning bumping areas on the second trace. In certain examples, the first and second impedance compensation traces may maintain a particular impedance and return loss along the first and second traces and may provide improved crosstalk performance between the differential pair and a nearby

differential pair as compared to previous impedance compensation schemes.

(54) FIG. **7** illustrates a generalized embodiment of an information handling system **700**. For purpose of this disclosure an information handling system can include any instrumentality or aggregate of instrumentalities operable to compute, classify, process, transmit, receive, retrieve, originate, switch, store, display, manifest, detect, record, reproduce, handle, or utilize any form of information, intelligence, or data for business, scientific, control, entertainment, or other purposes. For example, information handling system **700** can be a personal computer, a laptop computer, a smart phone, a tablet device or other consumer electronic device, a network server, a network storage device, a switch router or other network communication device, or any other suitable device and may vary in size, shape, performance, functionality, and price. Further, information handling system **700** can include processing resources for executing machine-executable code, such as a central processing unit (CPU), a programmable logic array (PLA), an embedded device such as a System-on-a-Chip (SoC), or other control logic hardware. Information handling system **700** can also include one or more computer-readable medium for storing machine-executable code, such as software or data. Additional components of information handling system **700** can include one or more storage devices that can store machine-executable code, one or more communications ports for communicating with external devices, and various input and output (I/O) devices, such as a keyboard, a mouse, and a video display. Information handling system **700** can also include one or more buses operable to transmit information between the various hardware components.

(55) Information handling system **700** can include devices or modules that embody one or more of the devices or modules described below, and operates to perform one or more of the methods described below. Information handling system **700** includes a processors **702** and **704**, an input/output (I/O) interface **710**, memories **720** and **725**, a graphics interface **730**, a basic input and output system/universal extensible firmware interface (BIOS/UEFI) module **740**, a disk controller **750**, a hard disk drive (HDD) **754**, an optical disk drive (ODD) **756**, a disk emulator **760** connected to an external solid state drive (SSD) **762**, an I/O bridge **770**, one or more add-on resources **774**, a trusted platform module (TPM) **776**, a network interface **780**, a management device **790**, and a power supply **795**. Processors **702** and **704**, I/O interface **710**, memory **720**, graphics interface **730**, BIOS/UEFI module **740**, disk controller **750**, HDD **754**, ODD **756**, disk emulator **760**, SSD **762**, I/O bridge **770**, add-on resources **774**, TPM **776**, and network interface **780** operate together to provide a host environment of information handling system **700** that operates to provide the data processing functionality of the information handling system. The host environment operates to execute machine-executable code, including platform BIOS/UEFI code, device firmware, operating system code, applications, programs, and the like, to perform the data processing tasks associated with information handling system **700**.

(56) In the host environment, processor **702** is connected to I/O interface **710** via processor interface **706**, and processor **704** is connected to the I/O interface via processor interface **708**. Memory **720** is connected to processor **702** via a memory interface **722**. Memory **725** is connected to processor **704** via a memory interface **727**. Graphics interface **730** is connected to I/O interface **710** via a graphics interface **732**, and provides a video display output **736** to a video display **734**. In a particular embodiment, information handling system **700** includes separate memories that are dedicated to each of processors **702** and **704** via separate memory interfaces. An example of memories **720** and **730** include random access memory (RAM) such as static RAM (SRAM), dynamic RAM (DRAM), non-volatile RAM (NV-RAM), or the like, read only memory (ROM), another type of memory, or a combination thereof. BIOS/UEFI module **740**, disk controller **750**, and I/O bridge **770** are connected to I/O interface **710** via an I/O channel **712**. An example of I/O channel **712** includes a Peripheral Component Interconnect (PCI) interface, a PCI-Extended (PCI-X) interface, a high-speed PCI-Express (PCIe) interface, another industry standard or proprietary communication interface, or a combination thereof. I/O interface **710** can also include one or more other I/O interfaces, including an Industry Standard Architecture (ISA) interface, a Small Computer

Serial Interface (SCSI) interface, an Inter-Integrated Circuit (I2C) interface, a System Packet Interface (SPI), a Universal Serial Bus (USB), another interface, or a combination thereof. BIOS/UEFI module **740** includes BIOS/UEFI code operable to detect resources within information handling system **700**, to provide drivers for the resources, initialize the resources, and access the resources. BIOS/UEFI module **740** includes code that operates to detect resources within information handling system **700**, to provide drivers for the resources, to initialize the resources, and to access the resources.

(57) Disk controller **750** includes a disk interface **752** that connects the disk controller to HDD **754**, to ODD **756**, and to disk emulator **760**. An example of disk interface **752** includes an Integrated Drive Electronics (IDE) interface, an Advanced Technology Attachment (ATA) such as a parallel ATA (PATA) interface or a serial ATA (SATA) interface, a SCSI interface, a USB interface, a proprietary interface, or a combination thereof. Disk emulator **760** permits SSD **764** to be connected to information handling system **700** via an external interface **762**. An example of external interface **762** includes a USB interface, an IEEE 1394 (Firewire) interface, a proprietary interface, or a combination thereof. Alternatively, solid-state drive **764** can be disposed within information handling system **700**.

(58) I/O bridge **770** includes a peripheral interface **772** that connects the I/O bridge to add-on resource **774**, to TPM **776**, and to network interface **780**. Peripheral interface **772** can be the same type of interface as I/O channel **712**, or can be a different type of interface. As such, I/O bridge **770** extends the capacity of I/O channel **712** when peripheral interface **772** and the I/O channel are of the same type, and the I/O bridge translates information from a format suitable to the I/O channel to a format suitable to the peripheral channel **772** when they are of a different type. Add-on resource **774** can include a data storage system, an additional graphics interface, a network interface card (NIC), a sound/video processing card, another add-on resource, or a combination thereof. Add-on resource **774** can be on a main circuit board, on separate circuit board or add-in card disposed within information handling system **700**, a device that is external to the information handling system, or a combination thereof.

(59) Network interface **780** represents a NIC disposed within information handling system **700**, on a main circuit board of the information handling system, integrated onto another component such as I/O interface **710**, in another suitable location, or a combination thereof. Network interface device **780** includes network channels **782** and **784** that provide interfaces to devices that are external to information handling system **700**. In a particular embodiment, network channels **782** and **784** are of a different type than peripheral channel **772** and network interface **780** translates information from a format suitable to the peripheral channel to a format suitable to external devices. An example of network channels **782** and **784** includes InfiniBand channels, Fibre Channel channels, Gigabit Ethernet channels, proprietary channel architectures, or a combination thereof. Network channels **782** and **784** can be connected to external network resources (not illustrated). The network resource can include another information handling system, a data storage system, another network, a grid management system, another suitable resource, or a combination thereof.

(60) Management device **790** represents one or more processing devices, such as a dedicated baseboard management controller (BMC) System-on-a-Chip (SoC) device, one or more associated memory devices, one or more network interface devices, a complex programmable logic device (CPLD), and the like, that operate together to provide the management environment for information handling system **700**. In particular, management device **790** is connected to various components of the host environment via various internal communication interfaces, such as a Low Pin Count (LPC) interface, an Inter-Integrated-Circuit (I2C) interface, a PCIe interface, or the like, to provide an out-of-band (OOB) mechanism to retrieve information related to the operation of the host environment, to provide BIOS/UEFI or system firmware updates, to manage non-processing components of information handling system **700**, such as system cooling fans and power supplies.

Management device **790** can include a network connection to an external management system, and the management device can communicate with the management system to report status information for information handling system **700**, to receive BIOS/UEFI or system firmware updates, or to perform other task for managing and controlling the operation of information handling system **700**. Management device **790** can operate off of a separate power plane from the components of the host environment so that the management device receives power to manage information handling system **700** when the information handling system is otherwise shut down. An example of management device **790** include a commercially available BMC product or other device that operates in accordance with an Intelligent Platform Management Initiative (IPMI) specification, a Web Services Management (WSMan) interface, a Redfish Application Programming Interface (API), another Distributed Management Task Force (DMTF), or other management standard, and can include an Integrated Dell Remote Access Controller (iDRAC), an Embedded Controller (EC), or the like.
(61) Management device **790** may further include associated memory devices, logic devices, security devices, or the like, as needed or desired.
(62) Although only a few exemplary embodiments have been described in detail herein, those skilled in the art will readily appreciate that many modifications are possible in the exemplary embodiments without materially departing from the novel teachings and advantages of the embodiments of the present disclosure. Accordingly, all such modifications are intended to be included within the scope of the embodiments of the present disclosure as defined in the following claims. In the claims, means-plus-function clauses are intended to cover the structures described herein as performing the recited function and not only structural equivalents, but also equivalent structures.
(63) The above-disclosed subject matter is to be considered illustrative, and not restrictive, and the appended claims are intended to cover any and all such modifications, enhancements, and other embodiments that fall within the scope of the present invention. Thus, to the maximum extent allowed by law, the scope of the present invention is to be determined by the broadest permissible interpretation of the following claims and their equivalents, and shall not be restricted or limited by the foregoing detailed description.

Claims

1. An information handling system comprising: a first differential pair on a printed circuit board of the information handling system, wherein the first differential pair includes: first and second traces; a first set of impedance compensations routed only on an inner-side of the first trace, wherein a first width of the first trace is increased at locations of each of the first impedance compensations; and a second set of impedance compensations routed only on an inner-side of the second trace, wherein a second width of the second trace is increased at locations of each of the second impedance compensations, wherein the first and second impedance compensations are substantially aligned; a second differential pair on the printed circuit board, the second differential pair includes third and fourth traces, wherein a first distance separates the first and second differential pairs based on a distance between an outer-edge of the second trace at phase tuning bumping areas of the first differential pair and an outer-edge of the third trace at phase tuning bumping areas of the second differential pair; a third differential pair on the printed circuit board, the third differential pair includes fifth and sixth traces; and a fourth differential pair on the printed circuit board, the fourth differential pair includes seventh and eighth traces, wherein a second distance separates the third and fourth differential pair based on a distance between an outer-edge of the sixth trace at the phase tuning bumping areas of the third differential pair and an outer-edge of the seventh trace at the phase tuning bumping areas of the fourth differential pair, wherein the first distance is greater than the second distance.

2. The information handling system of claim 1, wherein the second differential pair includes: a third set of impedance compensations routed only on an inner-side of the third trace; and a fourth set of impedance compensations routed only on an inner-side of the fourth trace, wherein the third and fourth impedance compensations are substantially aligned.
3. The information handling system of claim 1, wherein the first distance affects a crosstalk performance of the first and second differential pairs.
4. The information handling system of claim 1, wherein the third differential pair includes: a fifth set of impedance compensations routed on an inner-side of the fifth trace; a sixth set of impedance compensations routed on an outer-side of the fifth trace, wherein the fifth and sixth impedance compensation are substantially aligned; a seventh set of impedance compensations routed on an inner-side of the sixth trace; and an eighth set of impedance compensations routed on an outer-side of the sixth trace, wherein the seventh and eighth impedance compensation are substantially aligned; wherein the fourth differential pair includes: a ninth set of impedance compensations routed on an inner-side of the seventh trace; a tenth set of impedance compensations routed on an outer-side of the seventh trace, wherein the ninth and tenth impedance compensations are substantially aligned; an eleventh set of impedance compensations routed on an inner-side of the eighth trace; and a twelfth set of impedance compensations routed on an outer-side of the eighth trace, wherein the eleventh and twelfth impedance compensations are substantially aligned.
5. The information handling system of claim 1, wherein based on the first distance being greater than the second distance, the first and second differential pairs have a greater crosstalk suppression as compared to the third and fourth differential pairs.
6. The information handling system of claim 1, wherein the first and second differential pairs have a substantially equal impedance compensation as compared to the third and fourth differential pairs.
7. An information handling system comprising: a first differential pair on a printed circuit board of the information handling system, wherein the first differential pair includes: first and second traces; a first set of impedance compensations routed only on an inner-side of the first trace, wherein the first impedance compensations are located on the inner-side of the first trace at phase tuning bumping areas of the first differential pair, wherein a first width of the first trace is increased at locations of each of the first impedance compensations; and a second set of impedance compensations routed only on an inner-side of the second trace, wherein the second impedance compensations are located on the inner-side of the second trace at the phase tuning bumping areas of the first differential pair, wherein a second width of the second trace is increased at locations of each of the second impedance compensations, wherein the first and second impedance compensations are substantially aligned; and a second differential pair on the printed circuit board, wherein the second differential pair includes: third and fourth traces; a third set of impedance compensations routed only on an inner-side of the third trace wherein the third impedance compensations are located on the inner-side of the third trace at phase tuning bumping areas of the second differential pair; and a fourth set of impedance compensations routed only on an inner-side of the fourth trace, wherein the fourth impedance compensations are located on the inner-side of the fourth trace at the phase tuning bumping areas of the second differential pair, wherein the third and fourth impedance compensations are substantially aligned, wherein a first distance separates the first and second differential pairs based on a distance between an outer-edge of the second trace at phase tuning bumping areas of the first differential pair and an outer-edge of the third trace at phase tuning bumping areas of the second differential pair; a third differential pair on the printed circuit board, the third differential pair includes fifth and sixth traces; and a fourth differential pair on the printed circuit board, the fourth differential pair includes seventh and eighth traces, wherein a second distance separates the third and fourth differential pair based on a distance between an outer-edge of the sixth trace at the phase tuning bumping areas of the third differential pair and an outer-edge of the seventh trace at the phase tuning bumping areas of the fourth differential pair,

wherein the first distance is greater than the second distance.
8. The information handling system of claim 7, wherein the first distance affects a crosstalk performance of the first and second differential pairs.
9. The information handling system of claim 7, wherein the third differential pair includes: a fifth set of impedance compensations routed on an inner-side of the fifth trace; a sixth set of impedance compensations routed on an outer-side of the fifth trace, wherein the fifth and sixth impedance compensations are substantially aligned; a seventh set of impedance compensations routed on an inner-side of the sixth trace; and an eighth set of impedance compensations routed on an outer-side of the sixth trace, wherein the seventh and eighth impedance compensations are substantially aligned; wherein the fourth differential pair includes: seventh and eighth traces; a ninth set of impedance compensations routed on an inner-side of the seventh trace; a tenth set of impedance compensations routed on an outer-side of the seventh trace, wherein the ninth and tenth impedance compensations are substantially aligned; an eleventh set of impedance compensations routed on an inner-side of the eighth trace; and a twelfth set of impedance compensations routed on an outer-side of the eighth trace, wherein the eleventh and twelfth impedance compensations are substantially aligned.
10. The information handling system of claim 7, wherein based on the first distance being greater than the second distance, the first and second differential pairs have a greater crosstalk suppression as compared to the third and fourth differential pairs.
11. The information handling system of claim 7, wherein the first and second differential pairs have a substantially equal impedance compensation as compared to the third and fourth differential pairs.
12. A method comprising: routing a first trace of a first differential pair on a printed circuit board of an information handling system; routing a second trace of the first differential pair on the printed circuit board of the information handling system; routing a first set of impedance compensations along the first trace, wherein the first impedance compensations are routed only on an inner-side of the first trace, wherein a first width of the first trace is increased at locations of each of the first impedance compensations; routing a second set of impedance compensations along the second trace, wherein the second impedance compensations are routed only on an inner-side of the second trace, wherein a second width of the second trace is increased at locations of each of the second impedance compensations; routing a third trace and a fourth trace of a second differential pair on a printed circuit board of an information handling system, wherein a first distance separates the first and second differential pairs based on a distance between an outer-edge of the second trace at phase tuning bumping areas of the first differential pair and an outer-edge of the third trace at phase tuning bumping areas of the second differential pair; routing a fifth trace and a sixth trace of a third differential pair on a printed circuit board of an information handling system; and routing a seventh trace and an eighth of a fourth differential pair on a printed circuit board of an information handling system, wherein a second distance separates the third and fourth differential pair based on a distance between an outer-edge of the sixth trace at the phase tuning bumping areas of the third differential pair and an outer-edge of the seventh trace at the phase tuning bumping areas of the fourth differential pair, wherein the first distance is greater than the second distance.
13. The method of claim 12, wherein the first impedance compensations are located on the inner-side of the first trace at phase tuning bumping areas of the first differential pair, and the second impedance compensations are located on the inner-side of the second trace at the phase tuning bumping areas of the first differential pair.
14. The method of claim 12, wherein based on the first distance being greater than the second distance, the first and second differential pairs have a greater crosstalk suppression as compared to the third and fourth differential pairs.
15. The method of claim 12, wherein the first and second differential pairs have a substantially equal impedance compensation as compared to the third and fourth differential pairs.

16. The method of claim 12, the first distance affects a crosstalk performance of the first and second differential pairs.
17. The method of claim 12, the second distance affects a crosstalk performance of the third and fourth differential pairs.